

Poornachandran A

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PROFESSIONAL SUMMARY

Computer and Communication Engineering student at Amrita Vishwa Vidyapeetham specializing in **AI systems**, **LLM applications**, **DevOps**, and scalable **backend architectures**. Experienced in production security hardening, containerized deployments, and building end-to-end **RAG** and **digital twin** systems deployed on cloud infrastructure.

EDUCATION

Amrita Vishwa Vidyapeetham

B.Tech in Computer and Communication Engineering

- CGPA: 8.49/10 | Coursework: Data Structures & Algorithms, Machine Learning, AI, Computer Networks, OS, DBMS

Coimbatore, India

Jul. 2024 – Present

EXPERIENCE

Platforms & DevOps Intern

Jun. 2026 – Present

Spaceborn Beyond Autonomous

Remote

- Managed **Docker** containerized deployment of a **PHP 8.2** / **Apache 2.4** web application on **Render** behind **Cloudflare** **CDN**; hardened server configuration via custom **security.conf** enforcing **HTTP security headers** (CSP, HSTS, X-Frame-Options, CORP) and suppressing Apache server tokens to eliminate information disclosure
- Executed a full **penetration test** across 6 security domains — remediating **40+** **vulnerabilities** spanning **authorization bypass**, **session fixation**, **SQL/NoSQL injection**, **rate limiting**, and **payment integrity**; implemented **Razorpay** replay attack prevention via **MongoDB** idempotency locks and enforced **CI/CD** pre-merge **CVE** auditing with **Composer** and **Git** version control

Web Development Intern

Dec. 2025 – Jan. 2026

ShadowFox

Remote

- Built and deployed a responsive portfolio web app using **React.js** and **Tailwind CSS** on **Vercel** with **Git**-based **CI/CD**; applied modular component architecture, **UI performance optimization**, and cross-browser compatibility testing

PROJECTS

Neural Search Engine | *Python, Qdrant, Groq, Next.js, Express, Docker, Vercel, Railway* | [Live](#) | [GitHub](#) Sep.–Oct. 2025

- Built a production-grade **Retrieval-Augmented Generation (RAG)** system with **hybrid vector retrieval** using **Qdrant**, **semantic embeddings**, **RRF fusion**, **FlashRank reranking**, and **LLM**-based answer synthesis for high-accuracy **multi-document reasoning**
- Engineered a **Map-Reduce** reasoning pipeline and automated **Python** ingestion for chunking and **vector indexing**; deployed **Next.js** frontend on **Vercel**, **Express** backend on **Railway** with **Docker** containerization and multi-session persistent query history

AI-Driven 5G Network Digital Twin | *Python, FastAPI, SimPy, Docker, AWS EC2, Vercel* | [Live](#) | [GitHub](#) Feb. 2026 – Present

- Designed a **Digital Twin** architecture using **SimPy** simulation with an **LSTM+XGBoost** ML ensemble and **PPO reinforcement learning** agent to model realistic **5G network** dynamics (latency, resource utilization, traffic demand) across multiple base stations
- Built a **Network State Model** with **FastAPI WebSocket** REST layer for **real-time** telemetry and **what-if scenario analysis**; deployed backend on **AWS EC2** with **Docker**, frontend on **Vercel** with structured logging and **load-balancing** recommendations

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C, SQL

AI / ML: LLMs, Retrieval-Augmented Generation (RAG), Vector Embeddings, Semantic Search, Prompt Engineering, Supervised ML, Reinforcement Learning

Frameworks & Libraries: FastAPI, Next.js, React.js, Node.js, Express, Tailwind CSS, SimPy, Flask

Databases & Vector Stores: Qdrant, MongoDB, PostgreSQL, MySQL, SQLite

DevOps & Platforms: Docker, Git, GitHub, AWS EC2, Vercel, Railway, Render, Cloudflare, Postman, REST APIs

CERTIFICATIONS

Supervised Machine Learning: Regression and Classification – DeepLearning.AI & Stanford (Coursera), Dec. 2025

Introduction to Software Engineering – IBM (Coursera), Dec. 2025